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CSE445 Project Update

Week 1 (Sep 21 – Sep 27):

In the first week, our teacher gave us an introduction to the course and discussed about the course project. We were instructed to form groups and decide on a project topic. I formed a group, but we couldn't decide on a project topic as most of them were already selected.

Week 2 (Sep 28 – Oct 4):

During the second week, I sat down with my group mates and discussed on the available projects. We discussed about various possible approaches and tried to understand each project and after several discussions, we finalized on "Real-Time Location Tracking of Moving Objects". We chose this topic because thought it was distinct from other projects and involves video processing which aligns with my interest in computer vision.

Week 3 (Oct 5 – Oct 11):

In this week, I started learning the basics that I need to implement this project as I had no prior experience with machine learning. I reviewed many online tutorials and YouTube videos to understand the objectives, requirements, and existing methods. After reviewing several object tracking implementations, I found that most of them were implemented using deep learning frameworks like YOLOv8 combined with OpenCV.

Week 4 (Oct 12 – Oct 18):

I decided to follow a YouTube tutorial to create a practical version of the project using existing deep learning method to understand how object detection and tracking systems work. It gave me an overall idea of what is expected from this project. Though this version worked successfully, I discussed with my group mates and decided to consult with our teacher to confirm if we can use deep learning frameworks in this project.

Week 5 (Oct 19 – Oct 25):

This week, I along with my group mates consulted with our teacher regarding whether deep learning models like YOLOv8 were allowed for this project. Our teacher advised us not to use deep learning and to avoid using external libraries as much as possible. Sir explained that with using libraries to make we can implement the project within 50-60 lines of code, which does not contribute to the learning process. We were instructed to implement the machine learning logic and algorithms manually and train the model on our own.

Week 6 (Oct 26 – Nov 1):

Since our dataset for the project consists of raw video data, we had concern as none of us had computer powerful enough to process video and train the model. It would be very time consuming to use our devices, so we decided to consult our teacher regarding this issue. After we explained our concern, our teacher permitted us to user pre-existing datasets to train our model as long as our algorithm was implemented manually using classical ML techniques.

Week 7 (Nov 2 – Present):

As our project requirements and constrains were finalized, I started planning the overall structure of our project. I planned on building a frontend as well for easy video uploading and result visualization, while focusing on a custom machine learning backend. We decided on using OpenCV for video processing and visualization and NumPy for mathematical operations and algorithm, and Streamlit for the user interface.

During this week, I continued learning about background subtraction, optical flow, and simple tracking algorithms like Kalman filters. Besides, I created detailed task list to organize our workflow. We decided to start the code implementation from next week after our mid-term.